

WONDER DAILY

Transistors

Make a little book



1 Fold it into 8 boxes. Open it flat.



2 Fold short end to short end. Cut the thick line. Open it.



3 Fold the long way, pictures out. Push the ends in.



4 Fold into a book. Page 1 goes on the front!



Before transistors

Old machines used hot glass tubes that burned out. In 1947, scientists built the first transistor from gold foil and crystal.



A bead with three legs

A black bead with three thin legs is a transistor. Billions of them work unseen inside the machines around us.



A little turn

Like a tap, a tiny signal on the middle leg opens a path, so a small push controls a much bigger flow.



Made from sand

Silicon is found in sand. A tiny sprinkle of other atoms lets it carry electricity only when told. It is a semiconductor.



Billions on a chip

A chip can hold billions of transistors, printed with light through stencils. Some steer electricity with an unseen push.



On and off

A switch is on or off, like a one or a zero. Eight switches can make 256 different patterns.